

10_Database_Architecture.md

Part 1 — Database Foundation

Document Information

Property	Value
Project	Aetheris Trade
Document	Database Architecture
Version	1.0.0
Status	Draft
Owner	Priyanshu Rajpurohit

Purpose

This document defines the complete database architecture for Aetheris Trade.

It specifies:

- Database philosophy
- Data model
- Core entities
- Relationships
- Scalability principles
- Security principles

This document serves as the single source of truth for backend data design.

Database Philosophy

The database should be:

- Secure
- Scalable
- Modular
- Maintainable
- Normalized
- High Performance

Design Principles

- Single Source of Truth
 - Minimize Data Duplication
 - Strong Data Integrity
 - Optimized Read Performance
 - Scalable Relationships
 - Clear Naming Standards
-

Database Technology

Primary Database

PostgreSQL

ORM

Prisma ORM

Authentication

Clerk

Cache (Future)

Redis

File Storage

Cloudinary / Supabase Storage

High Level Database Structure

```
graph TD
    Users --> Portfolio
    Portfolio --> subgraph " "
        direction LR
        A[ ] --- B[ ]
    end
```



Core Entities

User

Stores account information.

Portfolio

Stores user's virtual investment portfolio.

Holding

Stores currently owned stocks.

Transaction

Stores buy/sell history.

Stock

Stores listed companies.

Market Data

Stores latest stock prices and metrics.

Watchlist

Stores bookmarked stocks.

AI Conversation

Stores AI chat history.

Learning Progress

Stores completed lessons and quiz results.

Competition

Stores competition details.

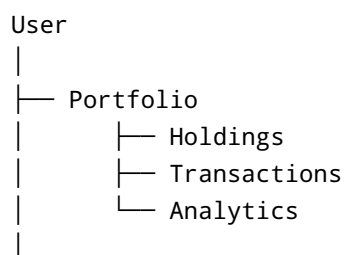
Leaderboard

Stores rankings.

Notification

Stores user notifications.

Entity Relationship Overview



```
├─ Watchlist
│
├─ AI Chats
│
├─ Learning Progress
│
└─ Notifications

Competition
│
└─ Leaderboard

Stock
│
├─ Market Data
└─ Historical Prices
```

Database Modules

Authentication Module

- Users
- Sessions

Market Module

- Stocks
- Market Data
- Historical Prices

Trading Module

- Portfolio
- Holdings
- Transactions

AI Module

- Conversations
 - Prompts
 - Responses
-

Learning Module

- Courses
 - Lessons
 - Progress
 - Quiz Results
-

Competition Module

- Competitions
 - Participants
 - Rankings
-

System Module

- Notifications
 - Activity Logs
 - Settings
-

Naming Convention

Tables

```
snake_case
```

Examples

```
users  
  
stock_prices  
  
portfolio_holdings  
  
trade_transactions
```

Columns

```
snake_case
```

Example

```
created_at  
  
updated_at  
  
user_id  
  
stock_symbol
```

Primary Keys

```
id
```

Foreign Keys

```
entity_id
```

Examples

```
user_id  
  
portfolio_id  
  
stock_id
```

Database Goals

- Fast reads
 - Consistent writes
 - Easy maintenance
 - Future scalability
 - Secure user data
 - AI-ready structure
-

Related Documents

- 08_Information_Architecture.md
- 09_Design_System.md
- 11_API_Architecture.md

Approval Status

Version: 1.0.0

Status: Draft

Next Document

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